1. Problem Statement

Big Mountain Ski Resort recently installed an additional chairlift to improve the distribution of visitors across the mountain. This new lift increases their operating costs by $1.54 million for the season. To offset these additional expenses, the resort is seeking strategic guidance on investment, optimizing ticket pricing, and maximizing the use of its facilities. With a goal of fully recovering the increased operational costs within the next two years, this report explores two key revenue-generating strategies:

1. **Implement a premium pricing strategy**, positioning the resort slightly above the average price point within its market segment.
2. **Increase revenue without altering ticket prices**, by reallocating resources, such as staff and operating hours, toward higher-margin facilities and services.

2. Data Analysis

2.1 Data Source and Cleaning

The datasets used for analysis include '**ski_data**' containing information from 330 resorts in the US that can be considered part of the same market share as Big Mountain Resort. Additional '**state_summary**' data containing population and area statistics for properly scaling ski data was acquired from Wikipedia.

There were originally 330 records and 27 fields (name, location, facilities, ticket price and operational information) in the ski dataset and there remain 277 records and 25 fields after cleaning. The fields removed include 'fastEight' and 'AdultWeekday' and the records removed include 2 for 'yearsOpen' entry being unreasonable and 47 (Weekend&Weekday) + 4 (Weekday) for not containing ticket price information.

The information on the resort of interest 'Big Mountain Resort' was found in 'Montana'. Additional information on population and area for each state was collected from Wikipedia and appended to the ski dataset. We have identified the **target feature - 'AdultWeekend'** - for our analysis.

2.1 Data Exploration and Feature Engineering

The 'state' dataset was used to assess whether all states could be treated equally in the modeling of ticket prices, or if there were patterns suggesting the need to group certain states. This dataset included seven numerical features, a couple of which were converted into density measures relative to area and population. Based on the results of principal component analysis (PCA) and average ticket prices by state, we found no compelling reason to treat any state differently (Fig. 1). After combining the 'ski_data' with 'state_summary' and scaling four features relative to state totals, we identified **28 numerical features** potentially influencing ticket prices. Ticket prices showed strong correlations with '*fastQuads*', '*Runs*', '*Snow_Making_ac*', and '*total_chairs*'. Additionally, the '*resort_night_skiing_state_ratio*' feature demonstrated a moderate correlation. (Fig. 2) While accurate modeling of ticket prices would require data on the number of visitors, our analysis suggests **there is potential to increase resort revenue by focusing on maximizing profits from 'fast quads' and further leveraging 'night skiing' capabilities.**

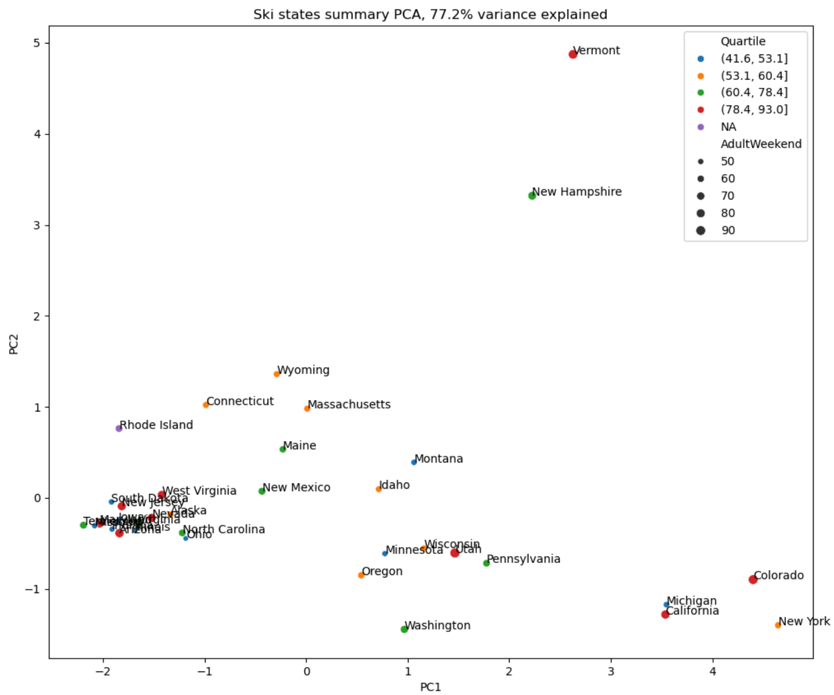

Figure 1. Two PCA components and average ticket prices by states

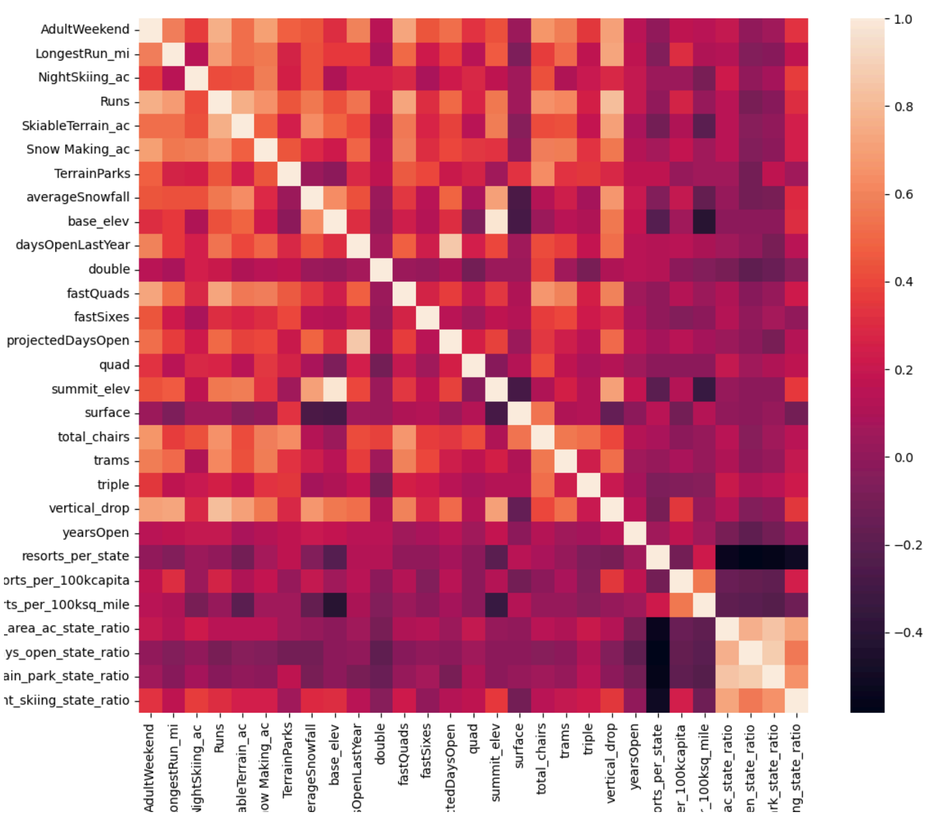

Figure 2. Correlations between features

3. Modeling

3.1 Evaluation Metric and Baseline Performance

We begin by calculating the average ticket price in the training dataset and use this value as a baseline for evaluating model performance. Specifically, we assessed R-squared, mean absolute error (MAE), and mean squared error (MSE), which resulted in approximately 0, **20**, and 600, respectively.

3.2 Selecting Best Model

As our first model, we selected a **linear regression** approach, fitting it to the scaled data. We then compared the results across several preprocessing variations. No significant difference was observed between the two imputation strategies tested,'median' and 'mean'. Using grid search cross-validation, we determined that the optimal number of features was 8 with 'vertical_drop' and 'Snow Making_ac' the largest two coefficients. The **best-performing linear regression model** using these features achieved an **MAE** of **10.5 ± 1.6** (1 standard deviation) in *cross-validation*. This result was consistent with its performance on the *test* split, which yielded an **MAE** of **11.8.**

For our second model, we selected a **random forest**. We performed hyperparameter tuning using grid search with cross-validation and found that the best-performing configuration used an ensemble size of 42, median imputation, and no data scaling. The most important features in the random forest model were found to be *'fastQuads'* and '*Runs*', outperforming the top two features identified in the linear regression model. The model achieved a **MAE** of **9.7 ± 1.4** in *cross-validation*, which was consistent with its performance on the *test* set - also yielding an **MAE** of **9.7**.

Given these results, we chose to proceed with the **random forest** model, as it demonstrated the best performance on the test data. This suggests stronger generalization capabilities and more accurate predictions on unseen data.

3.3 Scenario Modeling

The Big Mountain resort currently charges $81 per ticket, which is $20 less than the price predicted by our model. Even when adopting a conservative approach that accounts for the model's mean absolute error (MAE) of $10, the resort could reasonably increase its ticket price by $10, assuming no other unique factors are limiting its optimal pricing. To cover the additional cost of $1.54 million from installing a new chairlift, the ticket price would need to increase by $0.88 (=1540_000/(350_000*5)) per ticket. User sentiment is likely to be favorable, given the improvements made available to them.

Our model identified four dominant features influencing ticket prices: the number of **fast quads**, **total runs**, **snowmaking coverage**, and **vertical drop**. Big Mountain Resort already has a relatively large number of fast quads, 3. Other features, such as vertical drop, are inherent to the resort and cannot be altered without significantly affecting its branding, while expanding/reducing the snowmaking area is inefficient. The total number of runs turns out to be an effective approach to closing the gap between the 'current' and 'predicted' ticket prices for

Big Mountain. Fig. 3 tells us closing one run makes no difference. Closing 2 and 3 successively reduces support for ticket price and so revenue.

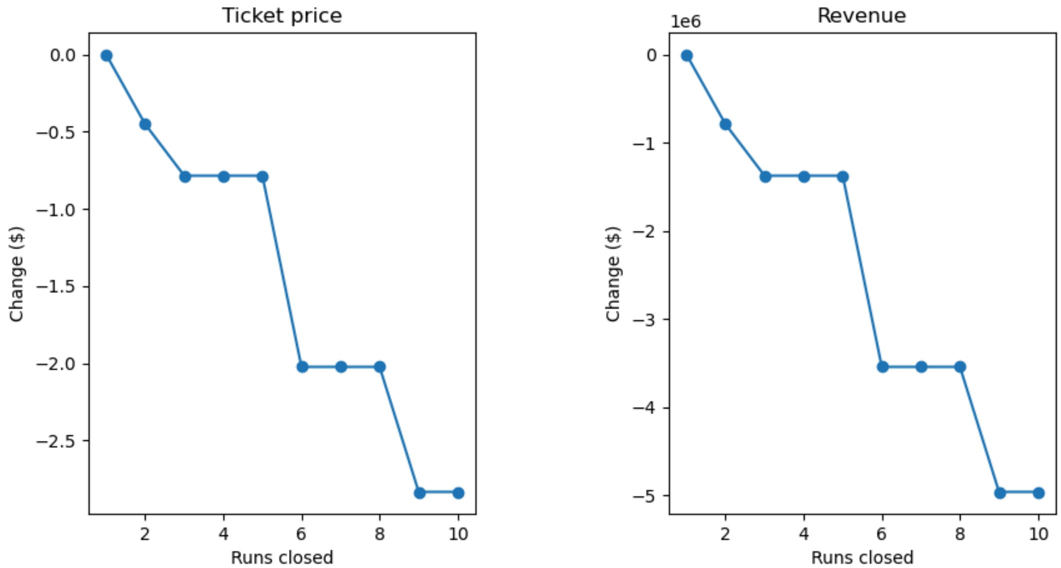

Figure 3. Change in (left) predicted ticket price and (right) revenue vs. Runs closed

4. Recommendations

As an immediate measure, we recommend **increasing the ticket price** by $1 at minimum. For future improvements, we recommend initially **closing a couple of runs for one year** to monitor changes in visitor numbers and net revenue. Based on the analysis, additional runs can be closed one at a time, continuing until net revenue growth rate plateaus. This approach will help the business align its pricing with market rates without incurring additional costs.

5. Future Work

We analyzed how adjustments to each of these features could impact pricing. However, any proposed changes, whether to ticket pricing or resort operations, must also consider their potential impact on visitor numbers and any associated costs.
Additional cost-related information, such as transportation options (availability of airports and highways), lodging, dining, and other amenities influencing overall affordability, would be valuable for a more comprehensive analysis. To better understand why our model predicted a significantly higher ticket price than what Big Mountain is currently charging, we would also need insights into their business model and profit margins.